

Total No. of Questions : 4]

SEAT No. :

PD224

[Total No. of Pages : 1

[6410]-553

T.E. (E&TC) (Insem)

HONORS IN ROBOTICS

Robot Programming & Simulation

(2019 Pattern) (Semester-II) (304183 HR)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume Suitable data, if necessary*

- Q1)** a) What is meant by ROS? Why it is good for ROBOTICS? [5]
b) What is difference between ROS and other meta Operating Systems? [5]
c) ROS for robotics software development: Pros and Cons? Explain. [5]

OR

- Q2)** a) Why should we use Robot OS? [5]
b) What are features of ROS? [5]
c) Write a short note on ROS framework: Operating system and its various releases? [5]

- Q3)** a) What are major components to be considered in Robot programming? [5]
b) What is Online and Offline robot Programing? [5]
c) Explain with typical Programming example such as Palletizing? [5]

OR

- Q4)** a) Explain Robot Programming technique as a Path in Space? [5]
b) What are various Textual Robot Languages? [5]
c) Explain with typical Programming Examples of Loading a Machine? [5]

